THE COMPARISON MAP ON EXT OF A TWO-TERM COMPLEX OVER THE QUOTIENT VANISHES AFTER A SHIFT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-local-cohomology/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The representation of a bounded complex over the quotient ring by a two-term complex whose first term is projective for the ideal power. Invoked inside the counted proof at source lines 5469-5476 and attributed to a lemma of another chapter (a cross-chapter pointer). Provides the explicit complex on which the comparison map is computed.
- The description of maps out of an almost free module and the factorisation lemma of this chapter. Both are invoked inside the counted proof (a cross-chapter pointer and the author's own lemma lemma-factor-hom). Convert the required vanishing of the comparison map into a factorisation statement, which is then verified diagrammatically.

Source reference. Original reference divisors-lemma-blowing-up-affine: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 4655–4655. Navigation pointer only: the counted text refers to the Stacks item divisors-lemma-blowing-up-affine, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-factor-hom: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 5486–5486. Navigation pointer only: the counted text refers to the Stacks item lemma-factor-hom, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

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Source reference. Original reference more-algebra-lemma-ext-1-annihilated-definite: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 5471–5471. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-ext-1-annihilated-definite, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference more-algebra-lemma-map-out-of-almost-free: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 5475–5475. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-map-out-of-almost-free, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *With $d = d(S)$ and $q_0 = q(S)$ as above. Then*

1. *for integers $n \geq c \geq 0$ with $n \geq \max(q_0 + (1+d)c, (2+d)c)$,*
2. *for K of $D(A/I^n)$ with $H^i(K) = 0$ for $i \neq -1, 0$ and $H^i(K)$ finite for $i = -1, 0$ such that $\mathrm{Ext}^1_{A/I^c}(K, N)$ is annihilated by I^c for all finite A/I^n-modules N*

the map

$$\mathrm{Ext}^1_{A/I^n}(K, I^n/I^{2n}) \longrightarrow \mathrm{Ext}^1_{A/I^n}(K, I^{n-(1+d)c}/I^{2n-2(1+d)c})$$

is zero.

Proof. The case $d > 0$. Let $K^{-1} \to K^0$ be a complex representing K as in More on Algebra, Lemma more-algebra-lemma-ext-1-annihilated-definite part (5) with respect to the ideal I^c/I^n in the ring A/I^n. In particular K^{-1} is I^c/I^n-projective as multiplication by elements of I^c/I^n even factor through K^0. By More on Algebra, Lemma more-algebra-lemma-map-out-of-almost-free part (1) we have

$$\mathrm{Ext}^1_{A/I^n}(K, I^n/I^{2n}) = \mathrm{Coker}(\mathrm{Hom}_{A/I^n}(K^0, I^n/I^{2n}) \to \mathrm{Hom}_{A/I^n}(K^{-1}, I^n/I^{2n}))$$

and similarly for other Ext groups. Hence any class ξ in $\mathrm{Ext}^1_{A/I^n}(K, I^n/I^{2n})$ comes from an element $\varphi \in \mathrm{Hom}_{A/I^n}(K^{-1}, I^n/I^{2n})$. Denote φ' the image of φ in $\mathrm{Hom}_{A/I^n}(K^{-1}, I^{n-(1+d)c}/I^{2n-(1+d)c})$. By Lemma lemma-factor-hom we can write $\varphi' = \sum a_i \psi_i$ with $a_i \in I^c$ and $\psi_i \in \mathrm{Hom}_{A/I^n}(M, I^{n-(2+d)c}/I^{2n-(2+d)c})$. Choose $h_i : K^0 \to K^{-1}$ such that $a_i \mathrm{id}_{K^{-1}} = h_i \circ d_K^{-1}$. Set $\psi = \sum \psi_i \circ h_i : K^0 \to I^{n-(2+d)c}/I^{2n-(2+d)c}$. Then $\varphi' = \psi \circ \mathrm{d}_K^{-1}$ and we conclude that ξ already maps to zero in $\mathrm{Ext}^1_{A/I^n}(K, I^{n-(1+d)c}/I^{2n-(1+d)c})$ and a fortiori in $\mathrm{Ext}^1_{A/I^n}(K, I^{n-(1+d)c}/I^{2n-2(1+d)c})$.

The case $d = 0$[1]. Let ξ and φ be as above. We consider the diagram

$$\begin{array}{ccccc} & & K^0 & & \\ & & \uparrow & & \\ K^{-1} & \xrightarrow{\varphi} & I^n/I^{2n} & \longrightarrow & I^{n-c}/I^{2n-c} \end{array}$$

[1]The argument given for $d > 0$ works but gives a slightly weaker result.

Pulling back to X and using the map $p^*(I^n/I^{2n}) \to \mathcal{O}_{Y_n}(n)$ we find a solid diagram

$$\begin{array}{ccccc} & & p^*K^0 & & \\ & & \uparrow & \searrow & \\ p^*K^{-1} & \longrightarrow & \mathcal{O}_{Y_n}(n) & \longrightarrow & \mathcal{O}_{Y_n}(n-c) \end{array}$$

We can cover X by affine opens $U = \mathrm{Spec}(B)$ such that there exists an $a \in I$ with the following property: $IB = aB$ and a is a nonzerodivisor on B. Namely, we can cover X by spectra of affine blowup algebras, see Divisors, Lemma divisors-lemma-blowing-up-affine. The restriction of $\mathcal{O}_{Y_n}(n) \to \mathcal{O}_{Y_n}(n-c)$ to U is isomorphic to the map of quasi-coherent $\mathcal{O}_U$-modules corresponding to the B-module map $a^c : B/a^nB \to B/a^nB$. Since $a^c : K^{-1} \to K^{-1}$ factors through K^0 we see that the dotted arrow exists over U. In other words, locally on X we can find the dotted arrow! Now the sheaf of dotted arrows fitting into the diagram is principal homogeneous under

$$\mathcal{F} = \mathcal{H}om_{\mathcal{O}_X}(\mathrm{Coker}(p^*K^{-1} \to p^*K^0), \mathcal{O}_{Y_n}(n-c))$$

which is a coherent $\mathcal{O}_X$-module. Hence the obstruction for finding the dotted arrow is an element of $H^1(X, \mathcal{F})$. This cohomology group is zero as $1 > d = 0$, see discussion following the definition of $d = d(S)$. This proves that we can find a dotted arrow $\psi : p^*K^0 \to \mathcal{O}_{Y_n}(n-c)$ fitting into the diagram. Since $n - c \geq q_0$ we find that ψ induces a map $K^0 \to I^{n-c}/I^{2n-c}$. Chasing the diagram we conclude that $\varphi' = \psi \circ \mathrm{d}_K^{-1}$ and the proof is finished as before. $\square$